

CHITA KADALA, Russian Federation

WMO: 307580

Lat: 52.026N

Lon: 113.306E

Elev: 693

StdP: 93.28

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.2	-35.1	-40.8	0.1	-37.2	-38.2	0.1	-34.8	9.8	-14.4	8.5	-14.0	0.2	60	0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	31.1	18.5	29.1	17.7	27.1	16.8	20.6	27.7	19.3	25.7	18.2	24.2	3.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	14.3	23.3	17.1	13.3	22.0	16.1	12.4	21.0	63.1	28.3	58.4	25.8	54.6	24.0	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.4	7.2	DB	-40.0	34.4	2.8	2.3	-42.0	36.0	-43.7	37.3	-45.3	38.6	-47.3	40.3	(4)
(5)			WB	-40.0	22.6	2.8	2.2	-42.1	24.1	-43.7	25.4	-45.3	26.6	-47.3	28.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-23.8	-17.8	-8.0	2.3	9.9	17.1	19.6	16.9	9.7	0.2	-12.2	-21.6	(6)	
	DBStd	15.89	6.02	6.17	6.24	4.66	4.20	3.65	2.80	3.17	3.98	4.79	6.81	6.62	(7)	
	HDD10.0	4676	1046	777	559	234	54	2	0	1	53	304	665	979	(8)	
	HDD18.3	6998	1304	1011	818	482	263	63	18	66	259	562	915	1237	(9)	
	CDD10.0	827	0	0	0	3	51	216	297	215	44	1	0	0	(10)	
	CDD18.3	107	0	0	0	0	1	27	57	21	1	0	0	0	(11)	
	CDH23.3	1646	0	0	0	2	79	523	714	297	30	0	0	0	(12)	
	CDH26.7	539	0	0	0	0	20	178	256	80	4	0	0	0	(13)	
	Wind	WSAvg	2.5	1.6	2.0	2.7	3.6	3.7	2.8	2.4	2.3	2.7	2.8	2.3	1.6	(14)
(15) Precipitation	PrecAvg	342	3	2	3	10	25	59	94	83	40	9	5	5	(15)	
	PrecMax	622	8	9	14	57	77	185	331	236	121	47	16	17	(16)	
	PrecMin	199	0	0	0	0	0	0	15	19	3	0	0	0	(17)	
	PrecStd	92	2	2	3	11	16	34	56	37	25	9	4	4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.4	1.6	12.0	21.8	29.4	34.0	34.9	31.2	26.9	17.6	6.8	-2.0	(19)	
		MCWB	-6.5	-2.1	4.5	9.4	15.9	19.3	21.4	18.6	15.2	8.6	2.0	-4.5	(20)	
	2%	DB	-8.9	-2.0	8.1	17.2	25.2	30.9	31.8	29.0	23.2	13.8	3.2	-5.2	(21)	
		MCWB	-10.4	-4.8	1.8	7.7	13.0	18.1	19.5	18.1	13.7	6.4	-0.4	-7.0	(22)	
	5%	DB	-11.8	-4.9	5.1	14.2	22.6	28.8	29.3	26.8	20.3	10.7	0.5	-8.0	(23)	
		MCWB	-13.0	-7.0	-0.1	6.1	11.4	17.2	18.2	17.4	12.2	4.5	-2.4	-9.4	(24)	
	10%	DB	-14.2	-7.4	2.2	11.6	19.9	26.3	27.3	24.4	17.9	8.0	-2.0	-11.0	(25)	
		MCWB	-15.2	-9.1	-2.0	4.5	10.0	15.8	17.5	16.5	11.2	3.0	-4.2	-12.1	(26)	
	0.4%	WB	-6.5	-2.3	4.9	10.6	17.0	21.5	23.6	21.5	16.9	9.1	2.4	-4.4	(27)	
		MCDB	-4.6	1.1	11.4	19.3	26.9	30.9	32.7	28.5	24.2	17.1	6.4	-2.2	(28)	
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-10.4	-4.9	2.0	8.2	14.2	19.9	21.2	19.7	15.0	6.7	-0.4	-7.1	(29)	
		MCDB	-9.0	-2.2	7.7	16.3	23.4	28.0	28.1	25.4	20.1	12.5	3.0	-5.4	(30)	
	5%	WB	-12.9	-7.0	0.0	6.4	12.2	18.1	19.8	18.4	13.5	4.9	-2.3	-9.5	(31)	
		MCDB	-11.8	-4.8	4.8	14.0	20.5	25.5	26.1	23.7	18.9	9.8	0.3	-8.0	(32)	
	10%	WB	-15.3	-9.3	-1.8	4.6	10.5	16.8	18.7	17.3	11.9	3.3	-4.2	-12.2	(33)	
		MCDB	-14.4	-7.6	2.1	11.1	18.6	24.1	24.6	22.6	17.0	7.5	-2.1	-11.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	15.6	15.3	14.1	15.1	15.0	13.5	13.0	14.0	12.6	11.6	11.2	(35)	
		MCDBR	12.7	16.8	17.3	19.3	21.0	19.9	18.1	17.6	18.9	16.8	12.8	12.1	(36)	
		MCWBR	11.6	14.4	12.2	12.0	11.2	9.4	8.0	8.4	10.3	10.3	10.0	10.6	(37)	
	5% WB	MCDBR	12.8	16.7	17.2	18.3	19.1	17.1	14.6	14.7	16.8	15.5	12.6	12.0	(38)	
		MCWBR	11.7	14.4	12.4	12.0	11.3	9.2	7.7	7.9	9.7	10.0	9.9	10.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.252	0.280	0.320	0.407	0.508	0.478	0.426	0.394	0.347	0.304	0.276	0.256	(40)	
	taud		2.208	2.173	2.155	2.065	1.834	2.031	2.252	2.323	2.394	2.425	2.285	2.167	(41)	
	Ebn at Noon		755	833	865	822	750	782	822	831	835	808	719	670	(42)	
	Edh at Noon		67	94	118	147	195	162	128	113	93	73	62	59	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.43	2.58	4.18	5.21	5.78	6.08	5.61	4.79	3.77	2.49	1.45	1.08	(44)	
	RadStd		0.04	0.06	0.14	0.37	0.36	0.43	0.47	0.32	0.28	0.16	0.08	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.98	-1.06	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page